

Software Requirement Specification

1. Introduction

1. Aim of the document

The document aims to present the software system PickyEaters.

Each of the following numbered subparagraph will point out a different characteristic of the system, starting from a general overview of the defined system, its hardware and software requirements, and its competitors – describing both the pros and cons of the system, compared to the system competitors.

Later in the document, each paragraph will present some functionalities of the system, starting with three user stories, then three functional requirements, and eventually the use case diagram.

2. Overview of the defined system

PickyEaters is a software system developed for all the picky eaters who struggle to find something to eat or drink when not eating at home.

The system expects three types of users:

- the base user, who will be called Pickie, interested in finding places to eat, by themselves or with their picky friends;
- the business owner user, who will be called Restaurateur, interested in promoting their business – a restaurant, pub, cafe, or similar;

The Pickie will be able to find places where they can find meals that suit their diet, by declaring ingredients they don't like and their dietary regimen (e.g.: halal, kosher, vegetarian, vegan, pregnant, etc...).

On the other hand, the Restaurateur shall register their business information into the system, providing also their menu, containing each dish and its ingredients. They are also responsible for maintaining updated menu information.

3. HW e SW requirements

The application is platform-independent and can be executed on any system where a Java Virtual Machine (JVM) is available. Two variants are provided: a GUI version, intended for systems with a graphical desktop environment, and a CLI version, which can be executed on headless systems without graphical capabilities.

The system can also run with two mode: a demo version where all the system will live in local RAM, so will be no persistency layer, and a full version that requires a PostgreSQL database (local or remote).

4. Related systems, Pros and Cons

PickyEaters is a unique system as it fills a gap in the food service market that is ignored by related systems, such as Tripadvisor, JustEat, and Glovo.

It is similar to Tripadvisor as its purpose is not to order and deliver food, but to show places where to eat out.

Yet it is also similar to JustEat and Glovo as it only concerns restaurants, pubs, cafes, and similar – not hotels, flights, etc... –, and offers an insight into a restaurant menu, showing dishes.

The potential of the idea behind PickyEaters cannot, unfortunately, be fulfilled because of the obvious lack of a corporate-like structure, funds, expertise, and time. This leads to a long list of cons that could, however, be easily mostly crossed out if only the system were developed in a proper environment; that list includes:

- Strong, prominent competitors;
- Simplistic and not captivating design of UIs;
- Simplistic functionalities;
- Underdevelopment of cybersecurity measures;
- Possibility of noncompliance with legal requirements;

As it comes to exploring the pros, it should be highlighted that:

- The system is wholly designed and developed ex novo: the software development process follows disciplined, uniform, and recent practices;
- There is room for adding, refining, and bettering functionalities;

- In the current market, where inclusion and customization are being more and more seen as priorities, no other similar system offers the possibility to customize and filter results at such a fine grain.

2. User Stories

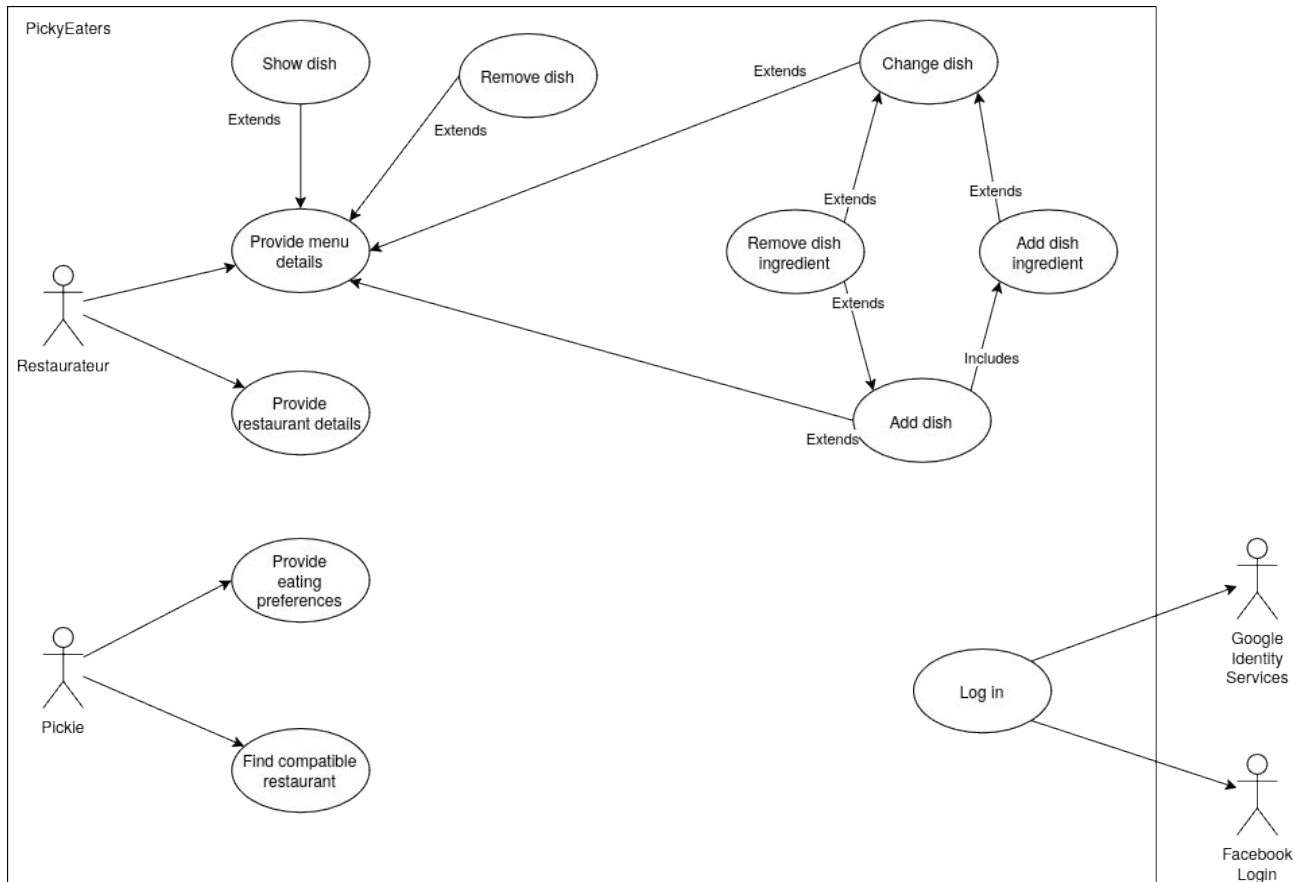
1. As a Pickie, I want to search restaurant without ingredients that I don't like, so that I can find a place to eat.
2. As a Restaurateur, I want to modify my menu (dish and ingredient of dish), so that Pickies can find my restaurant.
3. As a Restaurateur, I want to modify my restaurant details, so that Pickies can contact me.

3. Functional Requirements

1. The system shall provide the Pickie with the selection of disliked ingredients.
2. The system shall provide the Restaurateur with the selection of ingredients for their dishes.
3. The system shall provide the Pickie with a list of compatible restaurant based on the eating preference.

4. Use Cases

1. Overview Diagram



Notes:

1. All use cases, *Log in* excluded, include *Log in*.
2. The arrows from the actors *Pickie* and *Restaurateur* towards the use case *Log in* exist yet are not drawn for visual clarity.
3. Google Identity Service and Facebook Login will not be implemented in the first release.

2. Internal Steps

Name: **Add dish**

1. The Restaurateur requests to add a dish to the menu.
2. The System prepares a blank form.
3. The System fetches dish types.
4. The Restaurateur types the dish name and the dish description, one dish type, and adds an ingredient* or Remove an ingredient* per time.
5. For each ingredient selection, the System presents the ongoing list of selected ingredients and consequent allergens.
6. The Restaurateur indicates the operation is complete.
7. The System saves the dish.

8. The System display a success message.

Extensions:

3a. *Cannot fetch*: The System notifies the Restaurateur and terminates the use case.

5a. *Same ingredient selected more than once*: The System removes the double from the dish ingredient list.

6a. *No ingredient selected*: The System notifies the Restaurateur and require to add atleast an ingredient.

6b. *No name selected*: The System notifies the Restaurateur and require to insert dish name.

6c. *Duplicated name selected*: The System notifies the Restaurateur and require to insert alternative dish name.

6d. *No dish type selected*: The System notifies the Restaurateur and require to add a dish type.

7a. *Error while saving*: The System notifies the Restaurateur and terminates the use case.

Adds an ingredient*: It refers to the use case *Add ingredient*.

Remove an ingredient*: It refers to the use case *Remove ingredient*.

Storyboard

1. Add dish

 PickyEaters

Login

Le tue credenziali

Indirizzo email

luca.gasperini@pickyeaters.it

Password

Accedi ✓

 PickyEaters

Ciao, Luca Gasperini



Home
Cosa vuoi fare oggi?

Aggiorna
menù

Aggiorna
informazioni
del ristorante

 PickyEaters

Buongiorno, Luca Gasperini



< Indietro

Aggiorna menù
Il mio menù

+ Aggiungi piatto

Nome piatto	Descrizione piatto	Ingredienti	
Spaghetti alla carbonara	Spaghetti di grano 100% italiano trafilati al bronzo, croccante guanciale amatriciano d'allevamento biologico, avvolgente crema di uova laziali a KM 0, pecorino romano DOP ed aromatico pepe nero appena macinato.	Spaghetti di grano duro, Guanciale, Uova, Pecorino Romano DOP, Pepe nero, Sale	Modifica piatto Rimuovi piatto
Risotto alla crema di scampi	Riso Carnaroli selezionato, mantecato con una vellutata crema di scampi preparata con crostacei freschi, sfumatura di vino bianco, delicato fumetto di pesce, burro e Parmigiano Reggiano, impreziosito da polpa di scampi.	Riso Carnaroli, Scampi, Fumetto di pesce, Vino bianco, Burro, Parmigiano Reggiano, Olio extravergine di oliva, Prezzemolo, Sale, Pepe	Modifica piatto Rimuovi piatto
Sauté di cozze	Cozze fresche selezionate, saltate in padella con olio extravergine di oliva, aglio dorato, vino bianco, pomodorini freschi e prezzemolo appena tritato, servite con fragranti crostini di pane tostato.	Cozze, Olio extravergine di oliva, Aglio, Vino bianco, Pomodorini, Prezzemolo, Pane per crostini, Sale, Pepe	Modifica piatto Rimuovi piatto
Tiramisù	Soffici savoiardi imbevuti di caffè espresso, alternati a una vellutata crema al mascarpone con uova fresche e rifiniti con una generosa spolverata di cacao amaro.	Savoiardi, Mascarpone, Uova, Caffè espresso, Zucchero, Cacao amaro	Modifica piatto Rimuovi piatto

[< Indietro](#)

Aggiungi un piatto

Il mio piatto

Nome

Descrizione

Spaghetti di grano 100% italiano trafiletti al bronzo, croccante guanciale amatriciano d'allevamento biologico, avvolgente crema di uova laziali a KM 0, pecorino romano DOP ed aromatico pepe nero appena macinato.

Categoria

Bevanda 

Ingredienti















+ Aggiungi ingrediente

Allergeni

Salva piatto 

[< Indietro](#)

Aggiungi un piatto

Il mio piatto

Nome

Descrizione

Spaghetti di grano 100% italiano trafiletti al bronzo, croccante guanciale amatriciano d'allevamento biologico, avvolgente crema di uova laziali a KM 0, pecorino romano DOP ed aromatico pepe nero appena macinato.

Categoria

Bevanda 

Ingredienti







+ Aggiungi ingrediente

Allergeni

Salva piatto 

Piatto salvato
Spaghetti alla carbonara
è stato aggiunto al tuo menù

Continua

[< Indietro](#)

Aggiungi un piatto

Il mio piatto

Nome

Descrizione

Spaghetti di grano 100% italiano trafiletti al bronzo, croccante guanciale amatriciano d'allevamento biologico, avvolgente crema di uova laziali a KM 0, pecorino romano DOP ed aromatico pepe nero appena macinato.

Categoria

Bevanda 

Ingredienti







+ Aggiungi ingrediente

Allergeni

Salva piatto 

Piatto non salvato
Il piatto deve prevedere
un nome!

Continua

[< Indietro](#)

Aggiungi un piatto Il mio piatto

Nome

Descrizione

Spaghetti di grano 100% italiano trafilati al bronzo, croccante guanciale amatriciano d'allevamento biologico, avvolgente crema di uova laziali a KM 0, pecorino romano DOP ed aromatico pepe nero appena macinato.

Categoria

Piatto non salvato
Il piatto deve prevedere almeno
un ingrediente!

Allergeni

[< Indietro](#)

Aggiungi un piatto Il mio piatto

Nome

Descrizione

Spaghetti di grano 100% italiano trafilati al bronzo, croccante guanciale amatriciano d'allevamento biologico, avvolgente crema di uova laziali a KM 0, pecorino romano DOP ed aromatico pepe nero appena macinato.

Categoria

Piatto non salvato
Il piatto deve prevedere
una categoria!

Allergeni

[< Indietro](#)

Aggiungi un piatto Il mio piatto

Nome

Descrizione

Spaghetti di grano 100% italiano trafilati al bronzo, croccante guanciale amatriciano d'allevamento biologico, avvolgente crema di uova laziali a KM 0, pecorino romano DOP ed aromatico pepe nero appena macinato.

Categoria

Piatto non salvato
Spaghetti alla carbonara esiste
già nel menù!

Allergeni

[< Indietro](#)

Aggiungi un piatto Il mio piatto

Nome

Descrizione

Spaghetti di grano 100% italiano trafilati al bronzo, croccante guanciale amatriciano d'allevamento biologico, avvolgente crema di uova laziali a KM 0, pecorino romano DOP ed aromatico pepe nero appena macinato.

Categoria

Piatto non salvato
Errore tecnico durante il salvataggio. Riprova più tardi.

[Continua](#)

✕

✕

Allergeni

[Salva piatto ✓](#)
[< Indietro](#)

Aggiungi un piatto Il mio piatto

Nome

Descrizione

Categoria

Errore di caricamento
Errore tecnico durante il caricamento. Riprova più tardi.

[Continua](#)

Allergeni

[Salva piatto ✓](#)

2. Find restaurant



Login

Le tue credenziali

Indirizzo email
luca.gasperini@pickyeters.it

Password

Accedi ✓



Ciao, **Luca Gasperini**



Home

Cosa vuoi fare oggi?

Recensisci un
piatto

Trova un ristorante

Personalizza
le esigenze
alimentari



Buongiorno, **Luca Gasperini**

Esci

< Indietro

Trova un ristorante

I ristoranti compatibili con le mie esigenze

Città
Roma

Nome ristorante	Indirizzo	Numero di telefono	
Osteria del Gianicolo	Via del Gianicolo, 45 - 00165 Roma	+39 06 5123 4781	Vedi il menù
Trattoria Roma Antica	Via Appia Nuova, 218 - 00183 Roma	+39 06 5489 1623	Vedi il menù
Il Cortile Romano	Via Tuscolana, 134 - 00182 Roma	+39 06 6314 9057	Vedi il menù
La Terrazza del Tevere	Lungotevere degli Artigiani, 27 - 00153 Roma	+39 06 4278 3519	Vedi il menù
Sapori di Trastevere	Via della Lungaretta, 89 - 00153 Roma	+39 06 5891 2476	Vedi il menù
Bottega del Pantheon	Via dei Pastini, 41 - 00186 Roma	+39 06 6935 8142	Vedi il menù
Il Giardino di Testaccio	Via Marmorata, 76 - 00153 Roma	+39 06 5742 9368	Vedi il menù
Locanda Colle Oppio	Via Labicana, 112 - 00184 Roma	+39 06 7728 5401	Vedi il menù
La Forchetta d'Oro	Via Nomentana, 305 - 00162 Roma	+39 06 8619 7345	Vedi il menù

Buongiorno, **Luca Gasperini** Esci

< Indietro

Trova un ristorante
I ristoranti compatibili con le mie esigenze

Città

Roma

Nome ristorante

Indirizzo

Errore di caricamento

Errore tecnico durante il caricamento. Riprova più tardi.

Continua

Buongiorno, **Luca Gasperini** Esci

< Indietro

Trova un ristorante
I ristoranti compatibili con le mie esigenze

Città

Roma

Nome ristorante

Indirizzo

Errore nella ricerca

Errore tecnico durante la ricerca. Riprova più tardi.

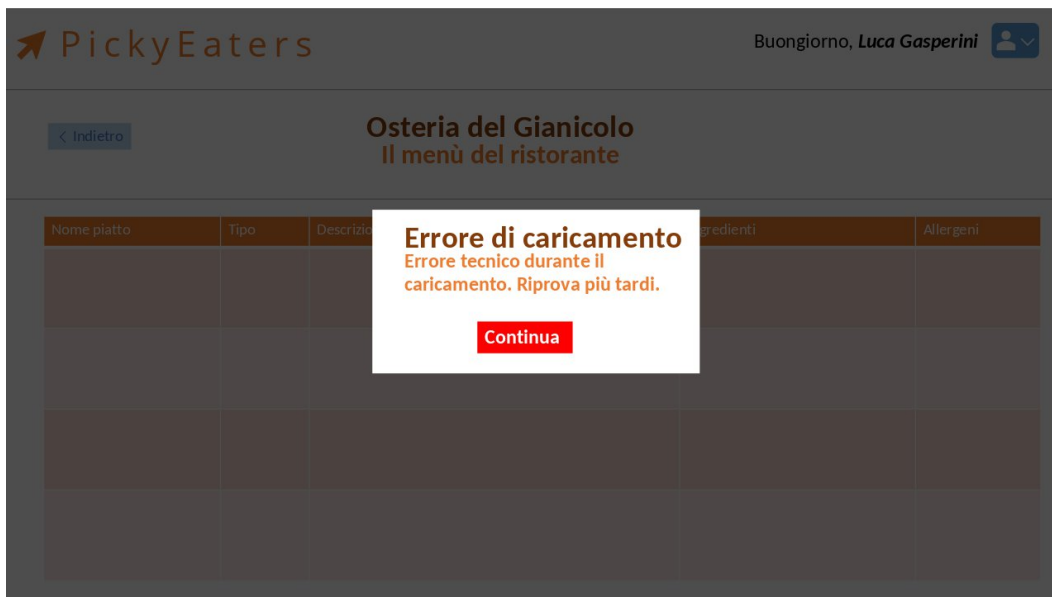
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Buongiorno, **Luca Gasperini**

< Indietro

Osteria del Gianicolo
Il menù del ristorante

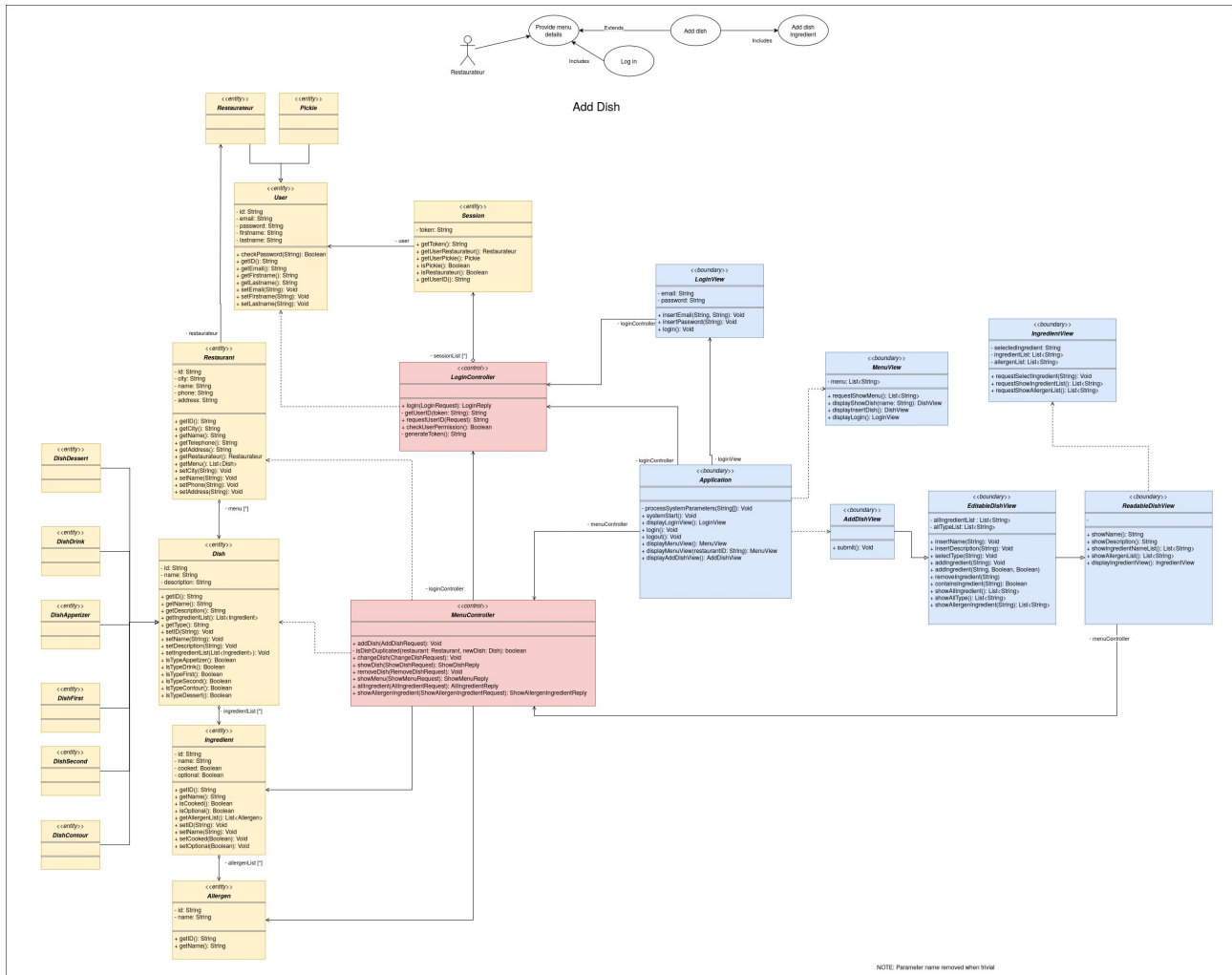
Nome piatto	Tipo	Descrizione	Ingredienti	Allergeni
Spaghetti alla carbonara	Primo piatto	Spaghetti di grano 100% italiano trafilati al bronzo, croccante guanciale amatriciano d'allevamento biologico, avvolgente crema di uova laziali a KM 0, pecorino romano DOP ed aromatico pepe nero appena macinato.	Spaghetti di grano duro, Guanciale, Uova, Pecorino Romano DOP, Pepe nero, Sale	Glutine, Uova, Latte
Risotto alla crema di scampi	Primo piatto	Riso Carnaroli selezionato, mantecato con una vellutata crema di scampi preparata con crostacei freschi, sfumatura di vino bianco, delicato fumetto di pesce, burro e Parmigiano Reggiano, impregniato da polpa di scampi.	Riso Carnaroli, Scampi, Fumetto di pesce, Vino bianco, Burro, Parmigiano Reggiano, Olio extravergine di oliva, Prezzemolo, Sale, Pepe	Crostacei, Latte
Sauté di cozze	Secondo piatto	Cozze fresche selezionate, saltate in padella con olio extravergine di oliva, aglio dorato, vino bianco, pomodorini freschi e prezzemolo appena tritato, servite con fragranti crostini di pane tostato.	Cozze, Olio extravergine di oliva, Aglio, Vino bianco, Pomodorini, Prezzemolo, Pane per crostini, Sale, Pepe	Molluschi, Glutine
Tiramisù	Dessert	Soffici savoiardi imbevuti di caffè espresso, alternati a una vellutata crema al mascarpone con uova fresche e rifiniti con una generosa spolverata di cacao amaro.	Savoiardi, Mascarpone, Uova, Caffè espresso, Zucchero, Cacao amaro	Glutine, Uova, Latte



Note:

1. Custom graphic for error message box will not be implemented in first release.
2. Orange box on fields will be not implemented in first release.
3. The welcome box on the application header will contain only the user's firstname.
4. Show menu restaurant in find restaurant will reuse show menu graphic of restaurateur in actual implementation.
5. All text displayed may vary depending on the actual implementation or language localization of the application.
6. Find restaurant form will contain dish type needed by the Pickie as a checkbox.

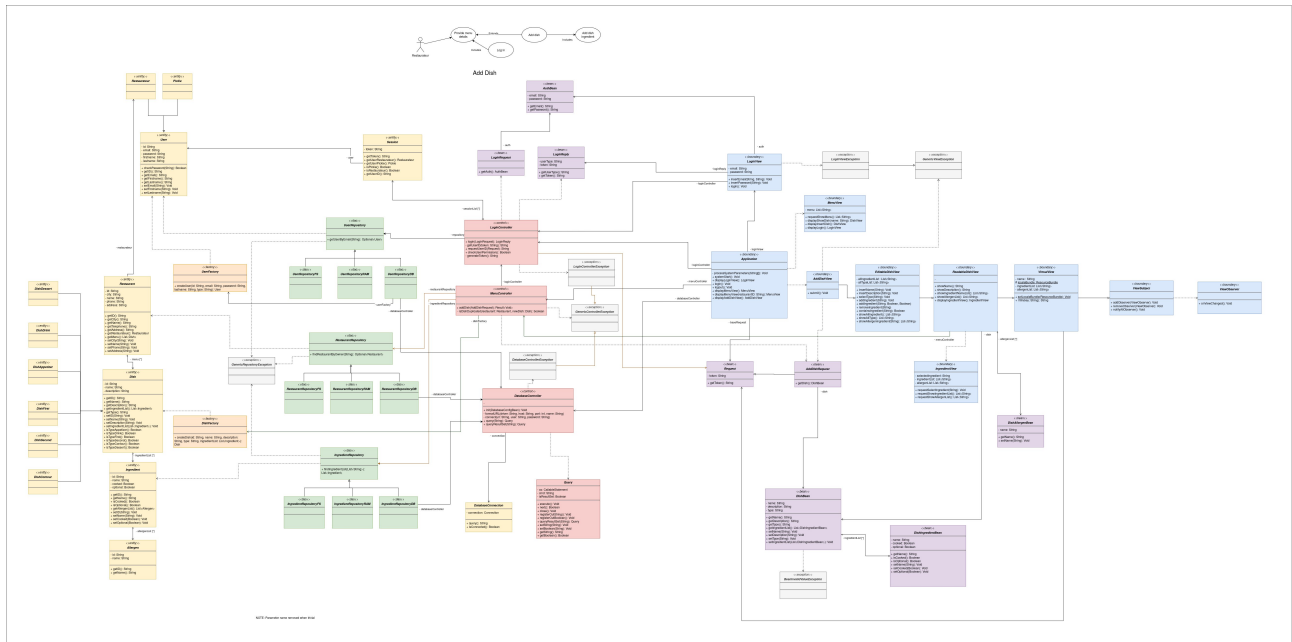
View Of Participating Classes



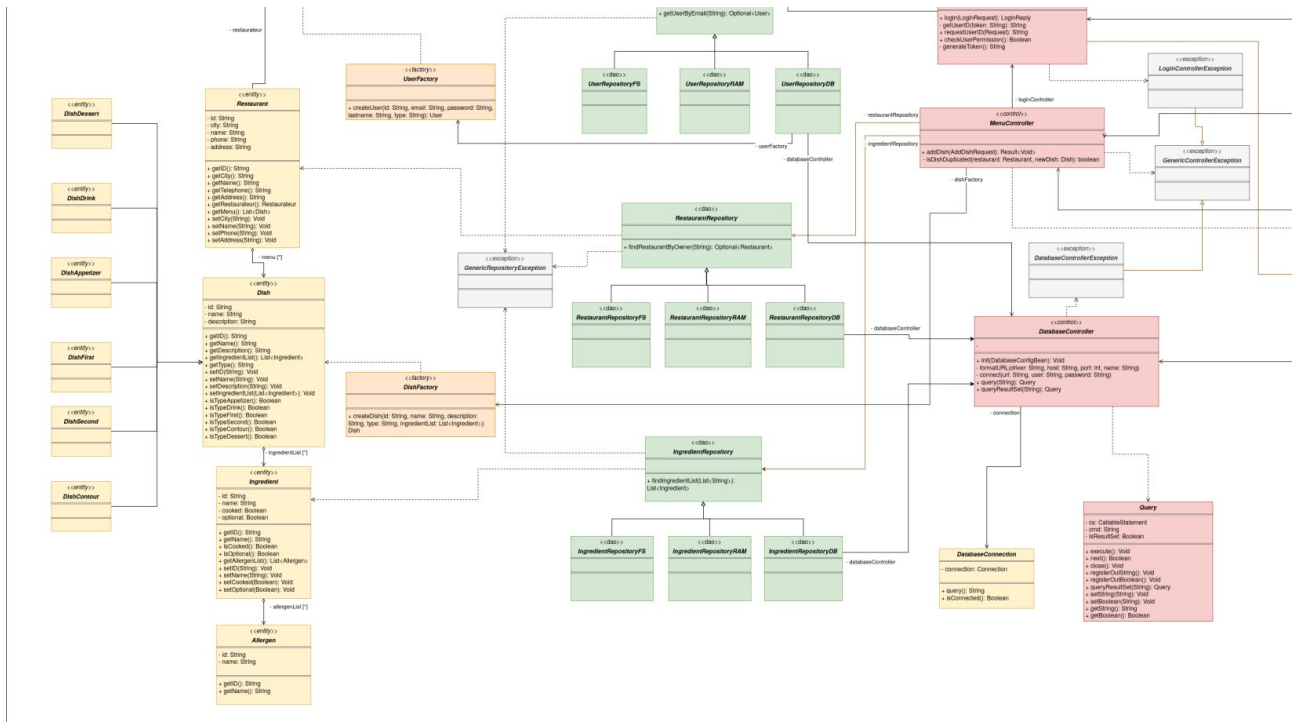
Note: The diagram represents a simplified version compared to the implemented code, so it's useful for visualizing the overall structure.

Class design

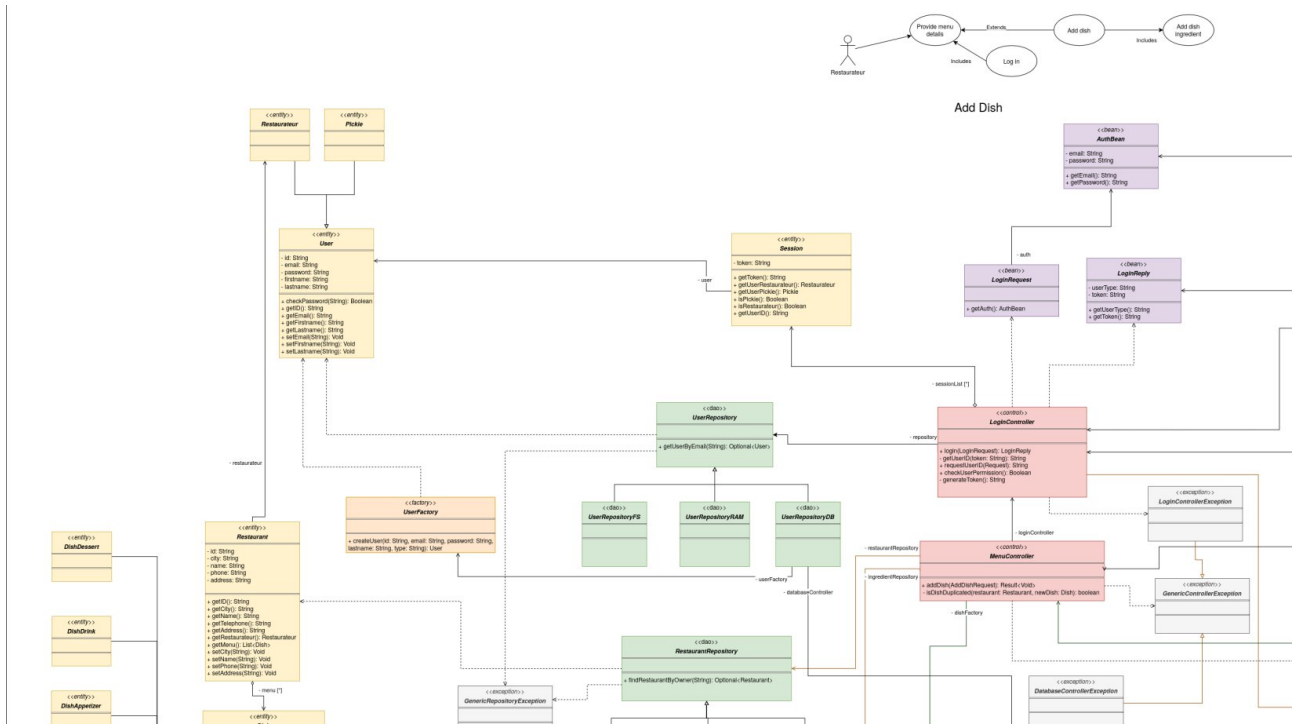
1. Full schema



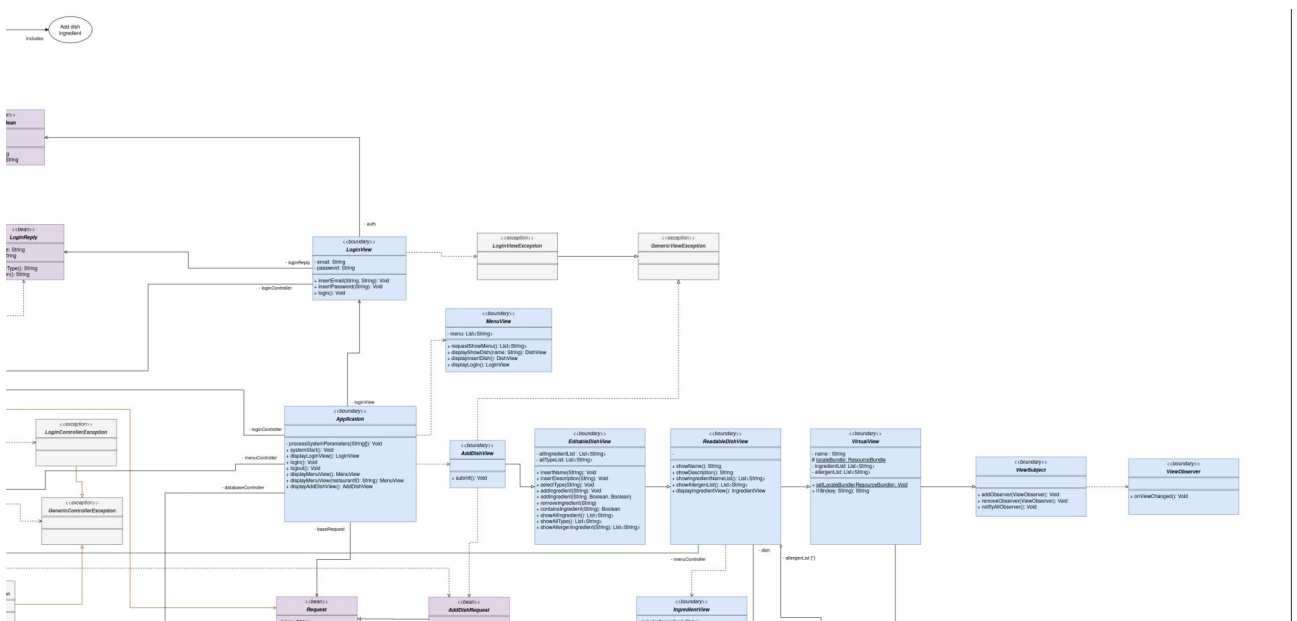
2. Bottom-left magnification (Model-Factory-DAO-Controller)



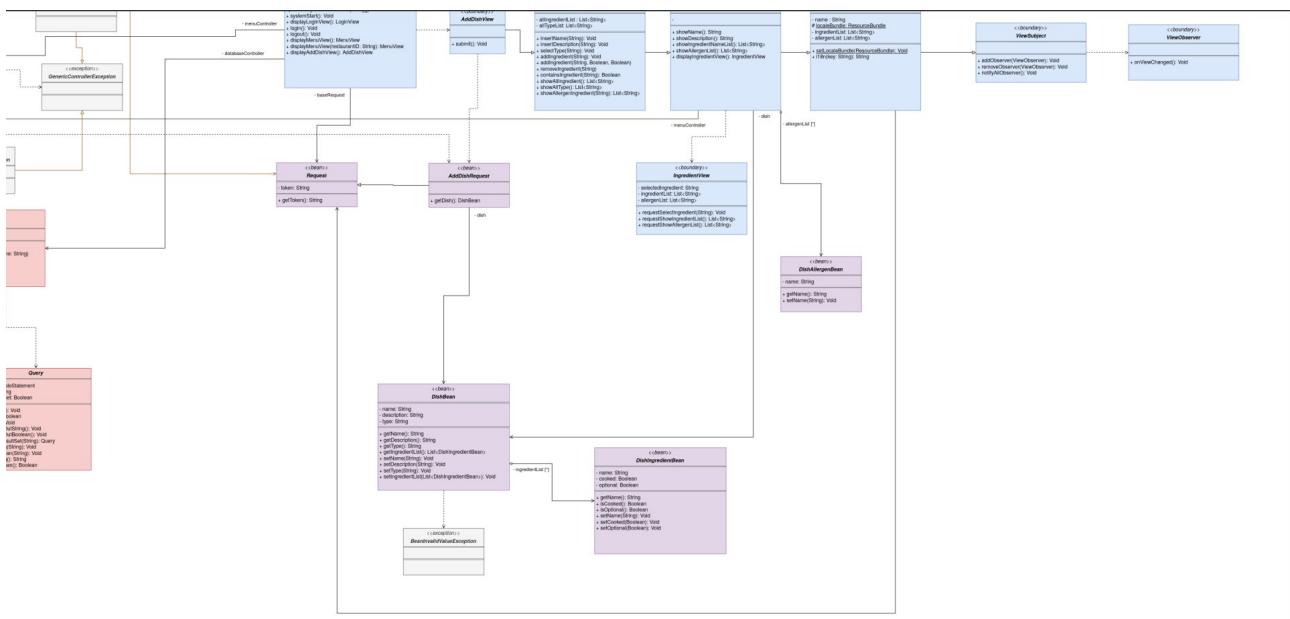
3. Top-left magnification (Model-Factory-DAO-Controller-Bean)



4. Top-right magnification (View-bean)



5. Bottom-right magnification (Controller-View-Bean)



Note 1: The diagram represents a simplified version compared to the implemented code, so it's useful for visualizing the overall structure.

Note 2: Repository are implemented as interface, so when they do not have the dependency, it's useful to indicate that the implementations of the interfaces depend on the class.

DESIGN PATTERNS

In the context of the system, the **Factory pattern** has been implemented to manage the creation of different types of objects in a centralized and structured way. The main purpose of this pattern is to separate the object creation logic from the classes that use those objects, reducing dependencies and improving the maintainability and extensibility of the system.

In particular, the **DishFactory** class has been implemented to handle the creation of the different types of Dish objects used within the application. Depending on the required category, the factory is responsible for creating the appropriate object without requiring the client classes to directly instantiate a specific implementation.

The DishFactory manages the creation of the following dish types: DishAppetizer, DishFirst, DishSecond, DishContour, DishDessert, and DishDrink. In this way, the logic related to the creation of dishes is centralized inside the factory, making it easier to introduce new

dish categories or modify the existing ones without significantly affecting the rest of the system.

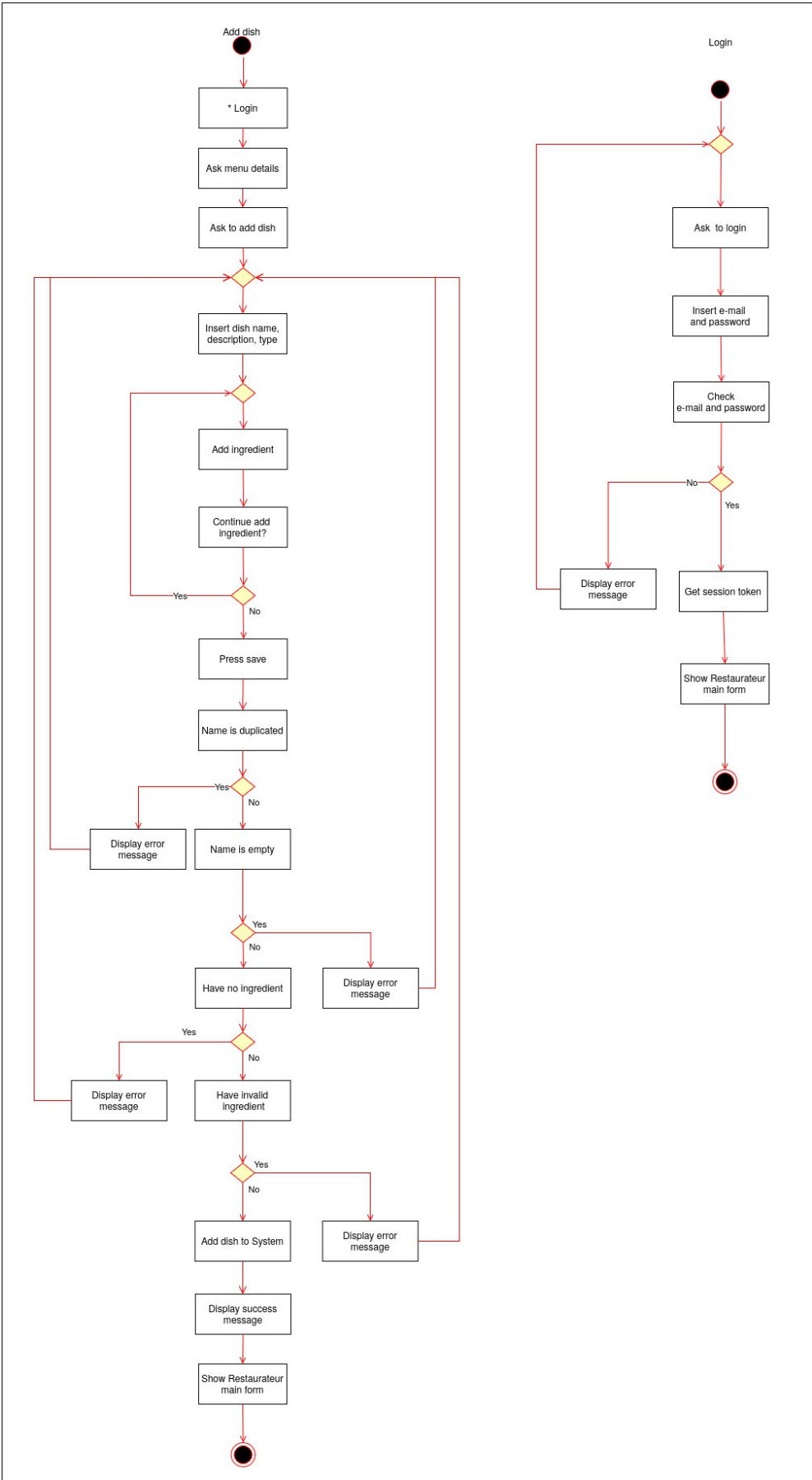
The same approach has been adopted for the management of users through the **UserFactory**. This factory is responsible for creating the appropriate User object according to the type of user required by the application. In particular, the system currently distinguishes between the Pickie and Restaurateur user types. Therefore, the client classes do not need to directly know which concrete user class has to be instantiated, since this responsibility is delegated to the UserFactory.

The use of these factories provides a clearer separation of responsibilities and avoids spreading object creation logic across different parts of the application. Furthermore, it makes the system more flexible, since additional types of dishes or users can be introduced by extending the corresponding factory logic.

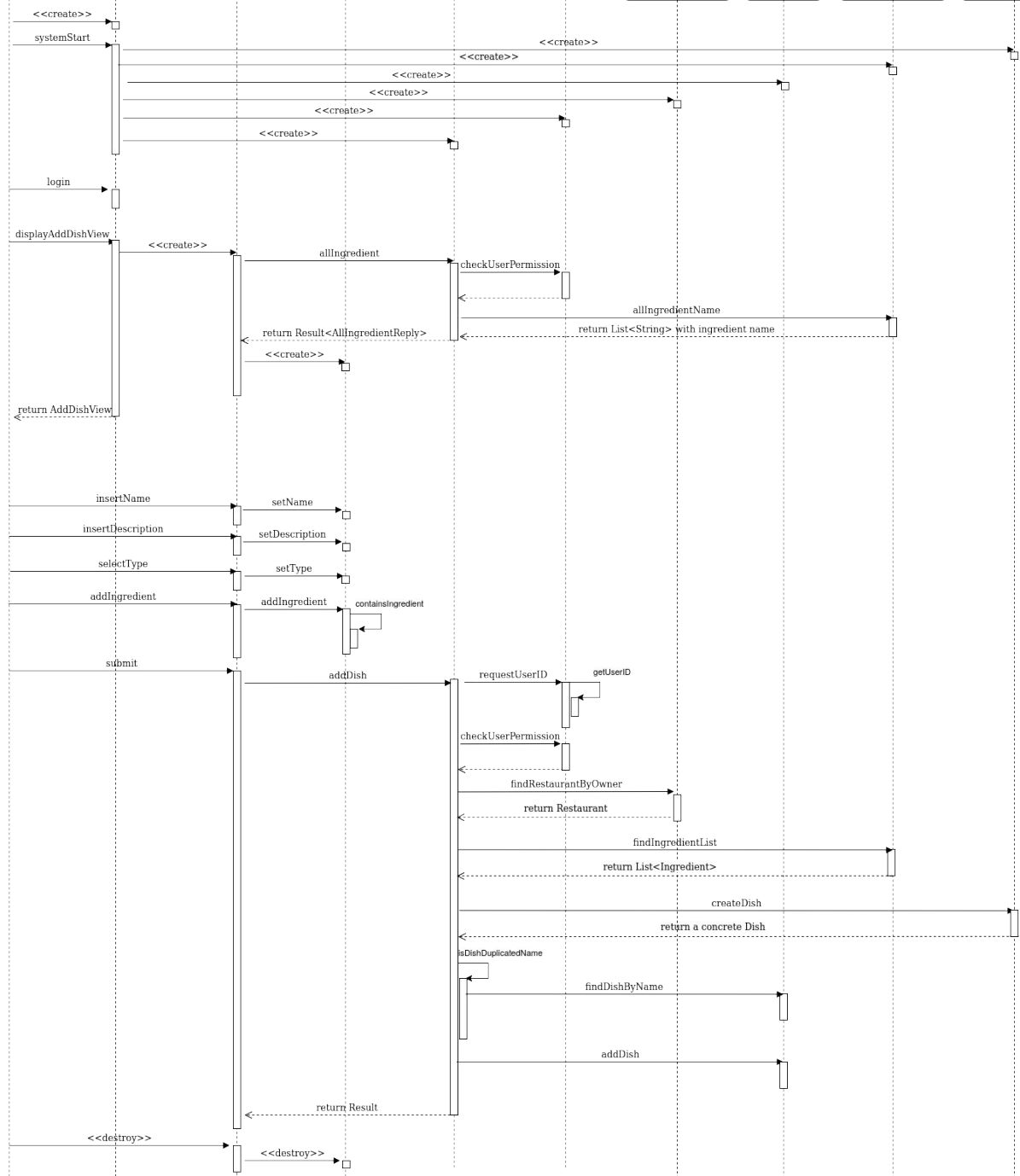
Moreover, other design patterns have been used in specific parts of the system. In particular, the **Observer pattern** has been adopted for the dynamic management of graphical interface elements in relation to the bean classes. Through this approach, the graphical components can react to changes in the state of the beans and update their content accordingly, without introducing strong dependencies between the graphical interface and the underlying data model.

Therefore, while the **Factory pattern** represents the main solution adopted for the creation of Dish and User objects through DishFactory and UserFactory, respectively, the use of the **Observer pattern** contributes to maintaining a more dynamic and loosely coupled interaction between the application beans and the graphical interface.

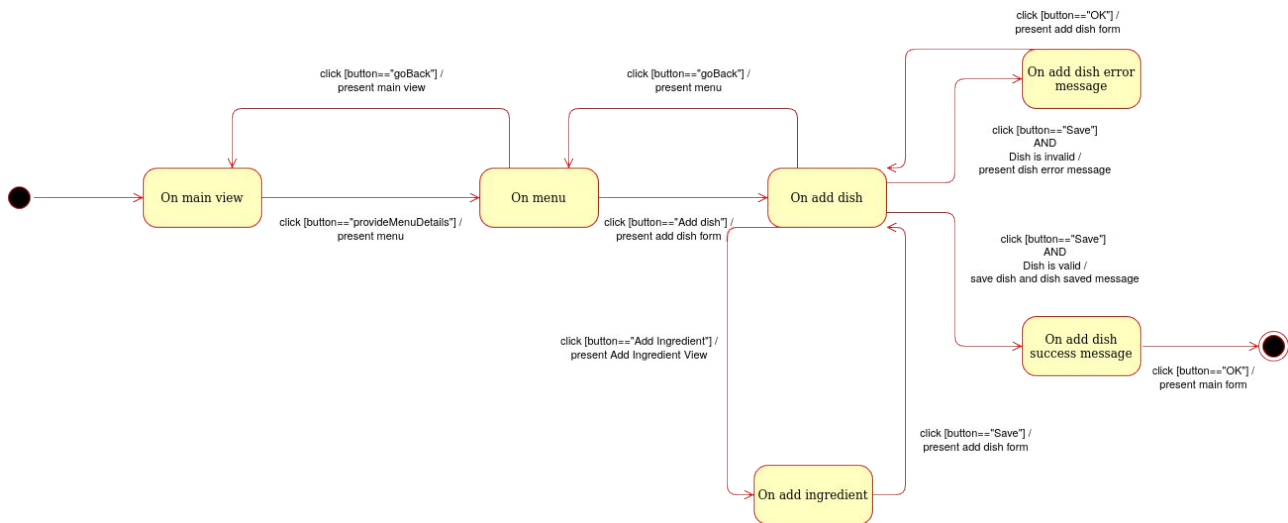
Activity Diagram



Note Actor	Note View	Note View	Note Bean	Note Controller	Note Controller	Note Repository	Note Repository	Note Repository	Note Factory
Restaurateur	Application	AddDishView	DishBean	MenuController	LoginController	RestaurantRepository	MenuRepository	IngredientRepository	DishFactory



State Diagram



Testing

For the testing phase, I have developed the following test classes, following the Use Case Diagram:

1. LoginFSTest
 - a. login(): used to test filesystem DAO
2. PickieTest
 - a. showEatingPreference()
 - b. editEatingPreference()
 - c. findRestaurant()
3. RestaurateurTest
 - a. showRestaurant()
 - b. editRestaurant()
 - c. showMenu()
 - d. addDish()
 - e. showDish()
 - f. changeDish()

Exception

There are various classes for exception handling in the project, the most important ones are:

1. GenericViewException
2. GenericBeanException
3. GenericControllerException
4. GenericRepositoryException
5. GenericFactoryException

They are, in fact, generic, and indicate an error at a certain level of the system, which may be passed on to the next higher level or resolved once it has been catch.

SonarQube

To check the correct behavior of this project, it was used SonarQube system. This system points out the possible bugs, possible vulnerabilities, and review that the functions are well implemented. You can check the results of the project in the following link:

https://sonarcloud.io/project/overview?id=lucagasperini_ispw

Public Git repository used:

<https://github.com/lucagasperini/ispw>

The video is on the documentation section of the repository.